

Atomic Layer Deposition of Aluminum Sulfide: Growth Mechanism and Electrochemical Evaluation in Lithium-ion Batteries

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ABSTRACT: This study describes the synthesis of aluminum sulfide (AlS_x) thin films by atomic layer deposition (ALD) using tris(dimethylamido)aluminum and hydrogen sulfide. We employed a suite of *in situ* measurement techniques to explore the ALD AlS_x growth mechanism including quartz crystal microbalance, quadrupole mass spectrometry, and Fourier transform infrared spectroscopy. A variety of *ex situ* characterization techniques were used to determine the growth characteristics, morphology, elemental composition, and crystallinity of the resultant AlS_x films. This study revealed that the AlS_x growth was self-limiting in the temperature range 100 - 250 °C, and the growth per cycle decreased linearly with increasing temperature from ~ 0.45 Å/cycle at 100 °C to ~ 0.1 Å/cycle at 250 °C. The AlS_x films were amorphous in this temperature range. We conducted electrochemical testing to evaluate the ALD AlS_x as a potential anode material for lithium-ion batteries (LIBs). The ALD AlS_x exhibited reliable cyclability over 60 discharge-charge cycles with a sustainable discharge capacity of 640 mAh/g at a current density of 100 mA/g in the voltage window of 0.6 – 3.5 V.

1. INTRODUCTION

Metal sulfides are versatile materials and can exhibit exceptional electrical, optical, magnetic, and mechanical properties¹. As a consequence, metal sulfides have been incorporated into a broad range of devices including heterogeneous catalysts²⁻³, energy conversion and energy storage devices^{1, 4-11}, transistors¹², photodetectors¹², and gas sensors¹³. Many of these applications require thin film coatings, and numerous coating techniques have been developed for depositing metal sulfide thin films such as chemical vapor deposition (CVD)¹³, physical vapor deposition (PVD)^{1, 14-16}, solution-based methods¹, and atomic layer deposition (ALD)¹⁷. Among these techniques, ALD has gained increasing recently as a means to deposit metal sulfide functional coatings due to its unique advantages in controlling materials growth fabricating nanostructures¹⁸⁻¹⁹. Utilizing alternating and self-limiting chemical reactions between gaseous precursors and a solid surface, ALD enables materials synthesis in an atomic layer-by-layer fashion. The self-limiting nature of ALD provides atomic-level precision over the film thickness and composition, and yields exceptionally uniform films over large areas and conformal coatings in complex geometries¹⁹⁻²². These benefits, as a result, have encouraged researchers to employ ALD coatings for energy devices^{10-11, 22-28}, catalysis²⁹, medical and biological devices³⁰⁻³¹, plasmonic devices³¹, nano- and micro-electromechanical systems³², and novel nanostructured materials³³. To date, metal sulfide ALD processes have been reported for ZnS³⁴⁻³⁵, CdS³⁶, CaS³⁷, BaS³⁸, SrS³⁸, Cu₂S^{8, 39}, In₂S₃⁴⁰, WS₂⁴¹⁻⁴², TiS₂⁴², PbS⁴³, SnS⁴⁴, GaS_x⁵⁻⁶, Li₂S⁴⁵, Co₉S₈⁴⁶, NiS_x⁴⁷, MnS⁹, MoS₂⁴⁸, Al₂S₃¹⁷, and their ternary derivatives^{7, 49-50}. Among metal sulfides, Al₂S₃ has been investigated electrochemically and showed great potential as an anode for lithium-ion batteries (LIBs)⁵¹, boasting a theoretical capacity above 1400 mAh/g (~4 times of the capacity of graphite anodes in commercial LIBs)⁵¹.

A previous study of Al_2S_3 ALD using trimethyl aluminum (TMA) and hydrogen sulfide (H_2S)¹⁷ reported that exceedingly long H_2S purge times of 110s were required in order to achieve a growth rate of 1.2 Å/cycle at 150 °C. The Al_2S_3 growth rate dropped monotonically with decreasing H_2S purge time and reached 0.35 Å/cycle at 40s. Typical ALD purge times are in the range of 1-10s. These very long purge times of 40 -110 s for Al_2S_3 ALD using TMA and H_2S are impractical, especially for applications requiring thicker films. This motivated us to evaluate a new process for aluminum sulfide ALD using tris(dimethylamido)aluminum (TDMA-Al) and H_2S as precursors. By performing *in situ* quartz crystal microbalance (QCM), quadrupole mass spectrometry (QMS), and Fourier transform infrared spectroscopy (FTIR) measurements, we explored the growth mechanism and surface chemistry for the AlS_x ALD. Next, we employed a series of *ex situ* characterization tools to analyze the ALD AlS_x films in order to determine their thickness, composition, and microstructure. After establishing this new ALD process, we proceeded to functionalize nitrogen-doped graphene nanosheets (NGNS) with ALD AlS_x to produce three-dimensional (3D) NGNS- AlS_x nanostructured composites. Finally, we evaluated the electrochemical performance of the 3D NGNS- AlS_x nanocomposites as an LIB anode material and measured the sustainable charge storage capacity and discharge-charge cyclability.

2. EXPERIMENTAL

The AlS_x ALD was performed in a custom viscous flow, hot-walled ALD reactor⁵² comprised of a 1 m stainless steel tube with an inner diameter of 5 cm. Before each deposition, substrates (i.e., Si(100) wafers and micromachined Si trenches) were first loaded in the reactor. During the ALD, a constant 300 sccm flow of ultrahigh purity Ar (UHP, 99.999%) passed through the flow tube at a pressure of ~ 1 Torr. The ALD AlS_x was performed using alternating exposures to TDMA-Al (Sigma-Aldrich,

USA) and Ar-balanced 1% H₂S (Matheson Tri-gas, USA) with Ar purge periods between each exposure. TDMA-Al normally exists as a dimer, Al₂(NMe₂)₆, but for simplicity we describe TDMA-Al in monomeric form, Al(NMe₂)₃. To provide sufficient vapor pressure, the solid Al(NMe₂)₃ was heated to 80 °C in a stainless steel reservoir, and 50 sccm UHP Ar was diverted through this reservoir during the Al(NMe₂)₃ exposures. This yielded a partial pressure of ~ 0.01 Torr Al(NMe₂)₃ in the flow tube. The 1% H₂S was stored in a pressure-regulated lecture bottle. A series of needle valves was used to deliver 1% H₂S pressure pulses of 0.2 Torr during the H₂S exposures. The AlS_x ALD timing can be described as: t₁-t₂-t₃-t₄, with t₁ and t₃ being the exposure times for the Al(NMe₂)₃ and H₂S, respectively, and t₂ and t₄ being the corresponding purge times, with all times in seconds (s).

The AlS_x ALD was systematically investigated using *in situ* QCM measurements. The QCM studies were conducted using a modified Maxtek Model BSH-150 sensor head and RC quartz crystal sensor (CNT06RCIA, Colnatec). The RC quartz sensors are less sensitive to temperature fluctuations compared to conventional AT quartz sensors, thereby reducing the effects of temperature-induced transients and temperature drift on the QCM measurements. The crystals were sealed within the sensor head using an electrically conducting high temperature epoxy (P1011, Epotek), and the sensor head was modified to provide backside purging of the crystal to prevent deposition on the back ^{6, 45,}

53.

Additional *in situ* studies were conducted using *in situ* quadrupole mass spectrometry (QMS) and Fourier transform infrared (FTIR) absorption spectroscopy measurements to explore the surface chemical reactions responsible for the AlS_x ALD. The QMS (Stanford Research Systems, Model RGA300) was located downstream from the sample position in a differentially pumped chamber

separated from the reactor by a 35 μm orifice. The FTIR (Nicolet 6700 FTIR spectrometer, Thermo Scientific) was operated in transmission mode in a separate ALD reactor as described previously ⁵⁴. The FTIR reactor utilized gate valves that were closed during the precursor exposures to prevent growth on the IR-transparent KBr windows. Substrates for FTIR measurements were prepared by pressing ZrO_2 nanopowder (Aldrich, < 100 nm TEM, > 25 m^2/g) into a stainless grid ^{55,56}. The grids were fabricated using photochemical machining (Fotofab, Inc.) and were 50 μm thick with 50 μm bars and 200 μm square openings. ZrO_2 is relatively transparent from 4000 to 800 cm^{-1} , the frequency range of interest for identifying surface functional groups, and provides a high surface area to increase the IR absorption signals. The nanopowder-filled grid was mounted onto a stage that could be heated to 500 $^\circ\text{C}$. This stage was then loaded into the FTIR reactor so that the IR beam passed through the center of the grid. During the *in situ* FTIR measurements, the substrate temperature was maintained at either 100 or 200 $^\circ\text{C}$ by the heated stage, and accordingly the reactor walls also were maintained at 100 or 200 $^\circ\text{C}$ to prevent precursor condensation.

For the *ex situ* analysis, ALD AlS_x films were deposited on Si wafers and micromachined Si trenches, and characterized using a variety of techniques. Aluminum sulfide is extremely sensitive to atmospheric moisture and must be kept in an inert environment at all times. To facilitate air-free transfer of the AlS_x films, an Ar-purged glove bag was installed over the exit flange on the ALD reactor. After removing the samples from the ALD reactor and into the glove bag, the samples were loaded into Ar-filled, sealed glass jars for transport. Spectroscopic ellipsometry (SE; alpha-SE, J. A. Woollam Co.) was employed and the ellipsometric data were fitted using a Cauchy model to determine the film thickness and refractive index. The ellipsometer was encased in an Ar-filled glove bag to prevent atmospheric exposure during these measurements. The film morphology was examined

using field emission scanning electron microscopy (FE-SEM, Hitachi S4700) equipped with energy dispersive X-ray spectroscopy (EDX). The crystallinity of the as-deposited and annealed AlS_x films were determined by X-ray diffraction (XRD, D8 Advance, Bruker). The film composition was measured by a Rutherford backscattering spectrometry (RBS: RBS400, CEA), and X-ray photoelectron spectroscopy (XPS: PHI Quantum 2000). The RBS and XPS measurements were conducted by Evans Analytical Group Laboratories (Sunnyvale, CA). The RBS measurements were performed using a 2.275 MeV He^{2+} ion beam with backscattering angle of 160° and a grazing angle of 100° . The XPS was equipped with a monochromatic $\text{Al-K}\alpha$ (1486.6 eV) X-ray source and depth profiling measurements were performed using Ar^+ sputtering. The sputter rate was calibrated using SiO_2 , accounting for a SiO_2 -equivalent rate of 7.81 Å/min. The analysis area was $1400 \times 300 \mu\text{m}$. The samples prepared for SEM, XPS, and RBS analysis were capped with ALD ZnS films to protect the ALD AlS_x films against air exposure, as successfully practiced in our previous study⁴⁵. The ALD ZnS coatings were deposited using alternating exposures to diethyl zinc (DEZ, Aldrich) and H_2S with the timing sequence 1-5-1-5 s.

In order to explore the potential for the ALD AlS_x to serve as a LIB anode, we deposited ALD AlS_x films on nitrogen-doped graphene nanosheet (NGNS) powders (ACS Material, USA) at 100°C and the resultant 3D NGNS- AlS_x nanocomposites were evaluated electrochemically. The NGNS powders have a surface area of $500 - 700 \text{ m}^2/\text{g}$ (determined by Brunauer-Emmett-Teller (BET) surface area analysis), consist of typically 1 – 5 atomic layer graphene nanosheets, and contain 1.0 – 3.0 at.% nitrogen and 7.0 – 7.5 at.% oxygen. After 50 cycles of ALD AlS_x using the timing sequence 60 – 60 – 120 – 60 on 59 mg NGNS powder, the resultant NGNS- AlS_x sample weighed 195 mg so that the AlS_x loading was $\sim 70 \text{ wt.}\%$. The NGNS- AlS_x was further mixed with Super P carbon black and

polyvinylidene fluoride (PVDF, Sigma-Aldrich) using a ratio of 8:1:1 and dissolved in an N-methyl-2-pyrrolidone (NMP, Sigma-Aldrich) solvent for fully homogenous mixing. The slurry was then cast onto a Cu foil and fashioned into a laminate using a 50 μm doctor blade. The laminate was dried in a furnace at 80 $^{\circ}\text{C}$ within an Ar-filled glove box for 24 hours. Next, the dried laminate was punched into 7/16 inch circular electrodes and subsequently assembled into CR2032 LIB coin cells in an Ar-filled glove box with H_2O and O_2 levels below 1 ppm. Li metal was used as the counter/reference electrode, a Celgard 2325 membrane was used as the separator, and 1 M lithium bis(trifluoromethanesulfonyl)imide (LITFSI, Sigma Aldrich) in 1,3-dioxolane (DOL, Sigma Aldrich) and 1,2-dimethoxyethane (DME, Sigma Aldrich) (DOL:DME = 1:1 by volume) was used as the electrolyte. The electrochemical discharge-charge testing was performed on an Arbin 4300 battery tester using either a voltage window of 0.01 - 3.0 V or a voltage window of 0.6 – 3.5 V for the $\text{NGNS-}\text{AlS}_x$ electrodes. In addition, control samples were assembled and tested following the same procedures described above, in which the ratio is 8:1:1 for active materials, Super P, and PVDF. The active materials in the control samples consisted of 70 wt.% micro-sized Al_2S_3 powder (Sigma-Aldrich) and 30 wt.% NGNS, and were mixed manually. All of the electrochemical testing was performed at room temperature.

3. RESULTS AND DISCUSSIONS

3.1. *In situ* QCM Measurements during AlS_x ALD

QCM is a powerful tool for evaluating new ALD processes^{6,45}, and can easily detect the mass changes associated with sub-monolayer quantities of material added or removed from the surface. [Figure 1\(a\)](#) shows the *in situ* QCM measurements of mass versus time recorded during 60 AlS_x ALD cycles with the timing sequence 5-5-10-5 s at 200 $^{\circ}\text{C}$. This timing sequence was established using *in situ* QCM

(see [Figures SI-1 in Supporting Information](#)) and ensures saturating $\text{Al}(\text{NMe}_2)_3$ and H_2S exposures. In order to establish a uniform starting surface for the AlS_x ALD, we deposited an ALD Al_2O_3 film on the QCM surface using alternating trimethylaluminum (TMA) and H_2O exposures with the timing sequence 1-5-1-5 s. [Figure 1\(a\)](#) reveals that the AlS_x ALD growth per cycle (GPC) was larger initially during the first ~ 20 ALD AlS_x cycles (0 - 500 s, highlighted orange area) on the Al_2O_3 surface before gradually decreasing and stabilizing to a constant growth per cycle value. The QCM data show a linear increase in mass versus time at $\sim 2.9 \text{ ng.cm}^{-2}.\text{cycle}^{-1}$ after ~ 500 s or ~ 20 ALD cycles. An enlarged view of four consecutive AlS_x ALD cycles in the stable growth regime is shown in [Figure 1\(b\)](#). The QCM mass is given by the blue trace in [Figure 1\(b\)](#), while the periods of $\text{Al}(\text{NMe}_2)_3$ and H_2S dosing are indicated by the black and blue traces, respectively. The $\text{Al}(\text{NMe}_2)_3$ exposures generate a mass increase $m_1 = \sim 4.7 \text{ ng.cm}^{-2}.\text{cycle}^{-1}$ and the H_2S exposures decrease the mass by $m_2 = \sim 1.8 \text{ ng.cm}^{-2}.\text{cycle}^{-1}$ so that the average net mass change $\Delta m = \sim 2.9 \text{ ng.cm}^{-2}.\text{cycle}^{-1}$. We can propose a mechanism for AlS_x ALD based on these mass changes:



in which “|” represents the surface and “g” indicates the gaseous phase. The Al compound reacts with y thiol (i.e., -SH) groups and releases y of the three dimethylamido- ligands as dimethylamine ($\text{HN}(\text{CH}_3)_2$) vapor ([Equation 1a](#)). Next, H_2S eliminates the remaining $(3 - y)$ dimethylamido- ligands as $\text{HN}(\text{CH}_3)_2$ forming stoichiometric Al_2S_3 and regenerating the surface thiols ([Equation 1b](#)). This mechanism makes several assumptions: (1) thiol groups are the reactive species for $\text{Al}(\text{NMe}_2)_3$ chemisorption; (2) the only gaseous product is $\text{HN}(\text{CH}_3)_2$; (3) the stoichiometry is Al_2S_3 . We will evaluate these assumptions below. Given [Equations 1a, 1b](#), and the atomic masses, we can define the QCM mass ratio as:

$$R = \Delta m/m_1 = 75/(159 - 45y) \quad (2)$$

The QCM data show $R = 0.57$ (Figure 1(b)). Consequently, $y = 0.85$, meaning that ~72% of the dimethylamido- ligands remain on the AlS_x surface following the Al precursor reaction (Equation 1a) at 200 °C. By repeating these measurements over the temperature range 100-225 °C (Figure SI-2a), we discovered that nearly 100% of the dimethylamido- ligands remain on the AlS_x surface following the Al precursor reaction at 100 and 125 °C. At temperatures ≥ 150 °C, ~70% of the dimethylamido- ligands remain after dosing the $\text{Al}(\text{NMe}_2)_3$.

Figure SI-2(a) shows in situ QCM measurements performed over the temperature range 100-225 °C to explore the effect of temperature on the AlS_x ALD. Assuming a density of 2.32 g/cm³ for bulk, crystalline Al_2S_3 , we calculated the GPC for ALD AlS_x using the QCM data in Figure SI-2(a), as indicated by the red symbols in Figure 2. To confirm the QCM data, we prepared ALD AlS_x films on Si(100) substrates using 50-600 ALD cycles at temperatures of 100, 150, 200, and 250 °C and measured the film thickness using ex situ spectroscopic ellipsometry (Figure SI-2(b)). Based on these data, we calculated the GPC at each growth temperature, and the results are given by the blue symbols in Figure 2. Both the data from QCM and ellipsometry in Figure 2 indicate that the ALD AlS_x GPC decreases with temperature. This trend is similar to our previous study of GaS_x ALD using $\text{Ga}_2(\text{NMe}_2)_6$ and H_2S ⁶. Comparing the two sets of GPC data in Figure 2, it is evident that the ellipsometry values are somewhat higher than the QCM values. This suggests that the density of the ALD AlS_x films is lower than the assumed, bulk value. Indeed, XRD measurements confirmed that the ALD AlS_x films are amorphous (Figure SI-3), and XPS elemental analysis detected residual C and N in the films (Figure 6), both of which will reduce the density of the ALD AlS_x films.

3.2. *In situ* FTIR and QMS Measurements during AlS_x ALD

Next, we conducted *in situ* measurements using FTIR and QMS to explore further the mechanism of AlS_x ALD. For the FTIR studies, we loaded a ZrO₂ nanopowder-filled grid into the ALD FTIR chamber and heated the ZrO₂ nanopowder to 400°C for 10 minutes. We then exposed the powder to oxygen (30 pulses with 350 sccm flow, 10 seconds each, 30 s purge between pulses). This treatment was performed to lower the FTIR background by oxidizing surface hydrocarbon contaminants. After cooling the powder to 200 or 100 °C, we performed 6 TMA/H₂O cycles for Al₂O₃ ALD (30-60-30-60s timing sequence) to establish a well-defined starting surface. The FTIR studies used 60s exposures and 60s purges for the AlS_x ALD, and used 100% H₂S. Twenty-four AlS_x ALD cycles were performed, and FTIR spectra were recorded after every precursor exposure. These ALD Al₂O₃ and AlS_x timing sequences were verified to saturate the nanopowder surfaces by extending the exposure times and observing no further changes in the FTIR signals. [Figure 3\(a\)](#) shows FTIR difference spectra after each precursor exposure for the 1st, 2nd, 3rd, 14th, and 18th AlS_x ALD cycles at 200°C. To generate these difference spectra, we subtracted the spectrum recorded during the previous precursor exposure. As a result, positive absorbance features indicate formation of new surface species, and negative features result from the removal of surface species. Positive features in the ranges of 2781 - 2929 and 889 - 1452 cm⁻¹ are seen after the first Al(NMe₂)₃ exposure. The 2781 - 2929 cm⁻¹ features are antisymmetric and symmetric C-H stretching modes⁵⁷⁻⁶⁰. The C-H stretching mode at 2781 cm⁻¹ is a Bolhmann band, and results from interaction between the N lone pair orbital and the C-H σ-orbital in N(CH₃)₂⁵⁹⁻⁶¹. This spectral feature indicates that some of the N(CH₃)₂ on the substrate surface are intact. We also see a negative absorbance at 3732 cm⁻¹ that we assign to the consumption of Al₂O₃ hydroxyl groups⁵⁹. Features in the range of 1106 - 1452 cm⁻¹ are mainly from CH₃ deformation and rocking modes⁵⁷⁻⁶⁰. In comparison to our previous study for ALD

GaS_x using Ga₂(NMe₂)₆ and H₂S⁶, the AlS_x ALD did not show any feature at 1575 cm⁻¹ that would indicate N=C double bonds from β-hydride elimination in the dimethylamido- ligand⁵⁹ (i.e. |N(CH₃)₂ → |N=CH₂ + CH₄ (g)). β-hydride elimination between neighboring methyls to release methane has been reported previously at temperatures above 200°C^{59, 57, 62,63}. We attribute the 889 to 1029 cm⁻¹ features to stretching modes of NC₂⁵⁷.

The first H₂S exposure produces negative absorbance features corresponding to the near-complete removal of CH₃ and NC₂. The difference spectra following consecutive Al(NMe₂)₃ and H₂S exposures are symmetric, indicating that ligand consumption and creation are comparable as expected for the proposed AlS_x ALD mechanism (Equations 1a and 1b). We also observed small changes in a feature at ~ 2400 cm⁻¹, that we assign to S-H stretching^{6, 64}. This S-H feature increases after H₂S exposures and decreases after Al(NMe₂)₃ exposures (Figure 3(a), inset). We integrated the NC₂ and S-H peaks after each precursor exposure, and these normalized values are shown in Figure 3(b). Figure 3(b) reveals that these spectral changes evolve over the first 5-7 AlS_x ALD cycles but then stabilize. We interpret this evolution as the gradual nucleation of the AlS_x film on the ALD Al₂O₃ surface, and this evolution stops once a continuous AlS_x film has formed. A similar nucleation behavior was observed during the QCM measurements of AlS_x ALD on Al₂O₃ (Figure 1). *In situ* FTIR measurements were also conducted at 100 °C (see Figure SI-4 in Supporting Information), and showed a similar ligand exchange as illustrated in Figure 3(a).

We performed *in situ* QMS measurements to identify and quantify the gas phase products of the AlS_x ALD surface reactions. First, we surveyed masses between m/z = 2 - 90 to identify the reaction products and found that dimethylamine (DMA, m/z = 45) was the only gaseous product. Figure 4

shows the DMA signals (blue trace) versus time recorded during AlS_x ALD at 200 °C using the sequence: 5-s Al(NMe₂)₃ dose, 50-s purge repeated 4 times, and then 10-s H₂S dose, 50-s purge repeated 4 times. In order for the QMS signals to stabilize after each exposure, we used longer purge times compared to the QCM measurements. Each precursor was dosed 4 times in succession so that the following three doses would reveal any possible background signals. These background signals are likely cracks of the parent compound formed by electron impact inside the QMS. In Figure 4, it is clear that the $m/z = 45$ intensity is lowest during the first Al(NMe₂)₃ dose, followed by a slight increase for doses two and three, and remaining constant during dose four. This pattern is quite reproducible between the three Al(NMe₂)₃ dosing sequences in Figure 4. We interpret the $m/z = 45$ signal during the third and fourth doses to be the background signal from Al(NMe₂)₃, and that the lower initial signal results from the depletion of Al(NMe₂)₃. This interpretation implies that little or no DMA product is generated (i.e. $y \sim 0$ in Equation 1a). In contrast, the four successive H₂S doses produce a monotonically decreasing $m/z = 45$ signal. It is possible that the H₂S reaction is not completely saturated in Figure 4, and that additional H₂S doses would reduce the $m/z = 45$ signal to zero. Our interpretation is that all of the $m/z = 45$ signal during the H₂S doses results from DMA formed by the ALD surface reactions. The fact that DMA persists even after four H₂S doses suggests that this reaction is relatively slow. The QMS measurements were repeated at a lower growth temperature of 100 °C (Figure SI-5) and the results were very similar with the exception that the DMA peak decreases more slowly during the H₂S dosing suggesting that the reaction is slower at 100 °C.

We assumed pure Al₂S₃ as the solid phase reaction product in Equations 1a and 1b. Based on the QMS data discussed above, however, impurities may result if the surface reactions fail to achieve completion due to slow kinetics and incomplete saturation. Some evidence for this is seen in the net

FTIR spectra recorded after each complete ALD cycle, (Figure SI-6). This figure shows a gradual increase in the NC₂ stretch, C-H bend, and C-H stretching regions with increasing ALD cycles consistent with the accumulation of DMA ligands in the film. Figure SI-7 shows the integrated absorbance of the 1450 cm⁻¹ C-H bending mode versus ALD cycles at 100 °C (black symbols) and 200 °C (red symbols). It is clear that the rate of impurity accumulation is ~2x lower at 200 °C.

3.3. Composition and Morphology of ALD AlS_x Films

We utilized RBS and XPS to analyze the composition of ALD AlS_x films deposited at 100 and 200 °C. Due to the air-sensitive nature of Al₂S₃, we applied an ALD ZnS capping layer over the AlS_x. ZnS was demonstrated in our previous study⁴⁵ to protect Li₂S from air exposure. The sample deposited at 100 °C consisted of a Si(100) substrate coated with 1250-cycle AlS_x and 1000-cycle ZnS, and the sample deposited at 200 °C was a Si(100) substrate coated with 1000-cycle AlS_x and 400-cycle ZnS. The two samples were stored under Ar prior to RBS characterization to minimize S exchange with atmospheric moisture. Figure 5 shows the RBS results of the two samples and more detailed results are described in Table 1. It is clear from Figure 5 that both samples comprise distinct ZnS and AlS_x layers. Figure 5(b) reveals that the AlS_x deposited at 200 °C is essentially pure, and yields an atomic ratio of S/Al ~1.5. In contrast, the AlS_x deposited at 100 °C (Figure 5(a)) shows very high C and N signals and a ratio of S/Al ~ 1.0. These findings are consistent with the FTIR measurements that showed a much larger accumulation of residual ligands at 100 °C (see Figure SI-6 and SI-7).

Table 1. Composition of ALD AlS_x and ZnS films determined using RBS measurements

Samples*	Layers		Atomic Concentration (at. %)						Atomic Ratio
			Zn	S	Al	N	C	Si	
	Layer 1	ZnS	48.6±1	51.4±1	--	--	--	--	ZnS _{1.06}
	Layer 2	AlS _x	--	23.2±1	22.4±1	22.3±5	32.1±5	--	AlS _{1.04}
	Bulk	Si	--	--	--	--	--	100±1	Si
	Layer 1	ZnS	49.1±0.5	50.9±1	--	--	--	--	ZnS _{1.04}
	Layer 2	AlS _x	--	60±1	40±1	--	--	--	Al ₂ S ₃
	Bulk	Si	--	--	--	--	--	100±1	Si

* Si(100) coated with 1,250-cycle AlS_x and 1000-cycle ZnS at 100 °C, while coated with 1,000-cycle AlS_x and 400-cycle ZnS at 200 °C.

We also applied XPS depth profiling analysis to the same two samples and the results are shown in [Figure 6](#). Consistent with the RBS and FTIR analyses, there are much more C and N impurities in the AlS_x layer deposited at 100 °C compared to 200 °C. Based on XPS analysis, the ratio of S/Zn is 0.68 at 100 °C and 0.85 at 200 °C, while the ratio of S/Al is 0.81 at 100 °C and 1.28 at 200 °C. The elemental ratios from XPS are not the same as those from RBS analysis, and this is probably due to the different element-specific sputtering rates for components in these films. The ratio of C/N is ~3 in these films, which is higher than the value of 2 expected for residual DME ligands, but this discrepancy might result from differential sputtering. To gauge the importance of the Ar storage used for these samples, we exposed one of the 100 °C films to air and performed XPS depth profiling. As shown in [Figure SI-8](#), the AlS_x layer has converted almost entirely to Al₂O₃ because of the air exposure, while the ZnS layer is unchanged.

High-resolution analysis of the Al XPS signals recorded from the AlS_x layers helped to determine

their chemical states (Figure SI-9). In the case of AlS_x deposited at 200 °C, the Al 2p peak is narrow and centered at a binding energy of 74.2 eV, as expected for Al_2S_3 ⁶⁵. In comparison, the AlS_x deposited 100 °C shows a peak at 73 eV and a shoulder at 73.8 eV. The higher binding energy peak is likely Al_2S_3 , while the lower energy peak is consistent with Al-N⁶⁶ and likely results from residual DMA ligands in the AlS_x film. Figure SI-9 also shows that, after oxidation, the Al 2p shifts to a higher binding energy of 76.5 eV, consistent with ALD Al_2O_3 ⁶⁷.

In order to examine the morphology and conformality of the ALD AlS_x films, we deposited a layer of ALD AlS_x using 1250 cycles onto trenched Si substrates, followed by a capping layer of ALD ZnS of 1000 cycles at 100 °C (see Figure 7). Figure 7(a) shows a cross-sectional view of the Si trenches, which are ~16 microns tall and separated by gaps of ~1.7 microns yielding an aspect ratio of ~9.4. Figure 7(b) and (c) show higher resolution cross-sectional images of the coated Si trenches at the top and bottom. The AlS_x and ZnS coatings are easily identified in these images, and are uniform and conformal along the trenched Si substrate. It is also disclosed that the GPC for AlS_x is ~0.4 Å/cycle which is consistent with ellipsometry (Figure 2), and for ZnS the GPC is 1.3 Å/cycle, very close to the value of 1.37 Å/cycle reported previously⁶⁸.

Based on all the information discussed above, the films deposited at 200 °C are composed primarily of Al_2S_3 with some DMA contamination. At lower growth temperatures (e.g., 100 °C), the surface reactions are sufficiently slow that they do not proceed to completion so that a significant fraction of the dimethylamido- ligands remain in the film where they substitute for S. Consequently, the S/Al ratio is less than 1.5. Based on Figures 4 and SI-5, it is likely that longer H_2S exposures would reduce the DMA content of the films.

3.4. Electrochemical Application as LIB Anodes

Al₂S₃ potentially is a very promising anode materials for LIBs, but to date only one study using micro-sized commercial Al₂S₃ has been published ⁵¹. The researchers showed that the micro-sized Al₂S₃ had severe capacity drop in 10 discharge-charge cycles and could not realize long-term cyclability. Using XRD and X-ray absorption spectroscopy (XAS), they disclosed that the electrochemistry of Al₂S₃ could be described as follows:



Equation 3 occurs at around 1.0 V, while Equation 4 proceeds at <0.5 V versus Li/Li⁺. Based on the above equations, Al₂S₃ should have a theoretical capacity of 1429 mAh/g, nearly 4 times higher than the widely used graphite anodes (372 mAh/g). Equation 3 is a conversion process, while Equation 4 is an alloying process. The two electrochemical reactions are supposed to be reversible during discharge-charge processes. However, Senoh et al ⁵¹ disclosed that the discharge capacity of the micro-sized Al₂S₃ faded quickly from ~1200 mAh/g for the 1st cycle to ~200 mAh/g for the 10th cycle at a current density of 100 mA/g. The underlying mechanism for the poor performance was believed to be structural irreversibility of the charging process or lithium extraction ⁵¹. Senoh et al found that the electrochemical reactions are reversible at the Al₂S₃ surface, but not reversible in the core of the Al₂S₃ microparticles. This suggests that the insulating nature and the micro-sized volume of the Al₂S₃ prevented reversible and stable electrochemical performance. In addition, the large volume changes during the micro-sized Al₂S₃ electrochemistry might lead to capacity loss.

Stimulated by these issues, we developed nanostructured, graphene-based AlS_x composites as LIB

anodes using ALD in order to achieve improved electrochemical performance. Figure 8 shows SEM images and EDX elemental maps for the nanostructured AlS_x based on NGNS. The pristine NGNS (Figure 8(a)) has very thin wrinkles of < 1 nm. After the ALD AlS_x coating of 50 cycles at 100 °C, the wrinkles of NSGS increased to ~ 15 nm, consistent with a GPC of 1.4 Å/cycle (see Figure 8(b)). Evidently, the growth of ALD AlS_x on NGNS is much higher than on Si wafers. This may result from a higher density of surface functional groups on NGNS or from CVD. Figure 8(b) also shows that the ALD coating is uniform all over the NGNS powder. This is further verified by EDX elemental mapping of carbon, aluminum, and sulfur shown in figure 8(c). The resulting NGNS- AlS_x nanocomposites have a loading of ~ 70 wt.% AlS_x . The NGNS- AlS_x nanocomposites were synthesized to address the issues encountered in micro-sized Al_2S_3 in the following ways: (1) ~ 7 nm thick nanofilms of AlS_x greatly shorten the diffusion pathways for lithium ions and electrons, (2) the NGNS substrates dramatically improve electrical conductivity, and (3) the porous NGNS provides a flexible scaffold to accommodate volume changes during cycling.

To verify the utility of our NGNS- AlS_x nanocomposites, we compared the NGNS- AlS_x performance with that of a mixture of 30 wt.% NGNS and 70 wt.% micro-sized Al_2S_3 during electrochemical cycling in the voltage window 0.01 - 3.0 V. In figure 9(a), the discharge capacity of commercial micro-sized Al_2S_3 continuously decreased from ~ 1200 mAh/g at the first cycle to 150 mAh/g at the 30th cycle. This is consistent with the study by Senoh et al ⁵¹. In contrast, the nanostructured NGNS- AlS_x exhibited a very impressive performance in the first 6 cycles: starting from a capacity of ~ 1000 mAh/g at the first cycle, then improved to ~ 1100 mAh/g, and remained until the 6th cycle. Beginning with the 7th cycle, however, the NGNS- AlS_x dropped in capacity continuously and remained only ~ 200 mAh/g at the 30th cycle. The Coulombic efficiencies (CE) of the nanostructured ALD AlS_x and

the commercial micro-sized Al_2S_3 all are lower than 100%, especially in the first few cycles where $\text{CE} < 70\%$ for the ALD AlS_x and $\text{CE} < 90\%$ for the micro-sized Al_2S_3 . These values indicate severe electrochemical irreversibility. Figure 9(b) illustrates the evolution of the electrochemical irreversibility of the NGNS- AlS_x nanocomposites with its discharge-charge profiles. Obviously, the discharge capacities are much higher than the charge capacities in all the cycles. In other words, after the cells were discharged, they could not be fully recharged. To examine the underlying mechanisms, we analyzed the dQ/dV profiles of the NGNS- AlS_x . Figure SI-10(a) shows that there is a reduction peak at around 0.6 V ascribed to the formation of solid electrolyte interphase (SEI) during the 1st discharge, and a strong reduction peak below 0.5 V related to lithium intercalation into NGNS and the formation of LiAl. In particular, the peaks at < 0.5 and 0 V were not sustainable and disappear by the 25th cycle. We postulate two reasons for the irreversibility of the NGNS- AlS_x nanocomposite and the NGNS- Al_2S_3 mixture. One is the formation of SEI, potentially increasing impedance and reducing capacity. Another reason is the formation of LiAl. Previously, Liu et al.⁶⁹ observed the formation of LiAl by *in situ* TEM and disclosed that the alloyed LiAl is prone to grow larger with increasing discharge-charge cycles, which impedes lithium extraction and the capacity quickly drops. In our study, the LiAl might have grown much larger after certain cycles and then isolated from further capacity contribution. This could be supported by the high capacity of 1100 mAh/g for the ALD NGNS- AlS_x in the first 6 cycles in figure 9(a). During the alloying of LiAl, volume changes can lead to poor electrochemical contact and isolation of active materials from further lithium extraction.

To address these issues, we repeated the cycling measurements using fresh coin cells in the potential window 0.6 – 3.5 V. This adjustment was based on three considerations: (1) SEI usually forms at < 0.6 V⁷⁰; (2) the alloying process of LiAl occurs at < 0.5 V⁵¹ with a volume change of $\sim 40\%$ compared

to the initial volume of Al_2S_3 ; and (3) excluding the capacity contribution from NGNS, for the lithium intercalation to NGNS occurs overwhelmingly at $< 0.3 \text{ V}$ ⁷⁰⁻⁷¹. We further extended the high limit to 3.5 V for a full exploration of capacity. Figure 9(c) shows the electrochemical cyclability and Coulombic efficiency of the ALD NGNS- AlS_x . We did not study further on the micro-sized Al_2S_3 with the new potential window, for it has been confirmed being inferior to the ALD nano-sized AlS_x . The ALD NGNS- AlS_x shows a discharge capacity of 800 mAh/g in the 1st cycle. After 60 discharge-charge cycles, the capacity is 640 mAh/g, which is 1.7 times the theoretical capacity of commercial graphite (372 mAh/g). In addition, the Coulombic efficiency is always $\sim 100\%$. Figure 9(d) shows the discharge-charge profiles during the first 5 cycles, clearly demonstrating reversible discharge and charge processes using the potential window of 0.6 – 3.5 V. Figure SI-10(b) also demonstrates that the electrochemical reactions are highly reversible.

4. CONCLUSIONS

Aluminum sulfide films were deposited by ALD using alternating exposures to $\text{Al}(\text{NMe}_2)_3$ and H_2S in the temperature range 100 - 250 °C. The films are smooth, amorphous, and conformal on high aspect ratio substrates. *In situ* QCM, QMS, and FTIR measurements revealed that the growth mechanism proceeds by ligand exchange reactions where a majority of the dimethylamino- groups are released during the H_2S exposures. The S/Al ratio and purity are highly dependent on the deposition temperature, and the ALD AlS_x films are very sensitive towards air exposure. The ALD AlS_x films exhibited excellent electrochemical performance as a LIB anode material in the potential window 0.6 – 3.5 V, showing high capacity and reliable cyclability.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website.

Additional figures

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Figure Captions:

Figure 1. *In situ* QCM measurements of ALD AlS_x at 200 °C using the timing sequence 5-5-10-5 s.

- (a) Mass of AlS_x film versus time during 60 ALD cycles beginning on ALD Al_2O_3 surface;
(b) Enlarged view of four consecutive ALD AlS_x cycles in the regime of constant growth per cycle. Lower traces indicate precursor pulsing during the alternating $\text{Al}(\text{NMe}_2)_3$ (red) and H_2S (black) exposures, and the m_1 , m_2 , and Δm are described in the text.

Figure 2. ALD AlS_x growth per cycle versus deposition temperature as measured by *ex situ* spectroscopic ellipsometry (solid symbols) and *in situ* QCM (open symbols).

Figure 3. (a) *In situ* FTIR difference spectra recorded after individual $\text{Al}(\text{NMe}_2)_3$ and H_2S exposures in the 1st, 2nd, 3rd, 14th, and 18th ALD cycles on an ALD Al_2O_3 surface at 200 °C. Right-hand spectra show expanded views of the S-H stretching region. (b) Integrated IR absorbance of the NC₂ region (red) and S-H region (blue) after each $\text{Al}(\text{NMe}_2)_3$ and H_2S exposure.

Figure 4. *In situ* QMS measurements at $m/e=45$ during AlS_x ALD at 200 °C where the $\text{Al}(\text{NMe}_2)_3$ H_2S exposures are separated into 4 separate micro-doses to evaluate the degree of saturation of the surface reactions. Lower traces indicate precursor pulsing during the $\text{Al}(\text{NMe}_2)_3$ (red) and H_2S (black) exposures as indicated.

Figure 5. RBS measurements of atomic concentration versus depth for films prepared at: (a) 100 °C using 1250 ALD AlS_x cycles on Si(100) and then capped with 1000 ALD ZnS cycles, and (b) 200 °C using 1000 ALD AlS_x cycles on Si(100) and then capped with 400 ALD ZnS

cycles.

Figure 6. Atomic concentration versus depth measured by XPS depth profiling analysis for films prepared at: (a) 100 °C using 1250 ALD AlS_x cycles on Si(100) and then capped with 1000 ALD ZnS cycles, and (b) 200 °C using 1000 ALD AlS_x cycles on Si(100) and then capped with 400 ALD ZnS cycles.

Figure 7. SEM images of ALD AlS_x film deposited in high aspect ratio Si trenched substrate using 1250 ALD AlS_x cycles and then capped with 1000 ALD ZnS cycles at 100 °C: (a) lower magnification image showing entire trenches covered by the double layers of AlS_x and ZnS, and higher magnification images showing (b) the top and (c) bottom of the cross sections of the coated trenches.

Figure 8. SEM images of (a) pristine and (b) ALD AlS_x coated NGNS, and (c) EDX elemental mapping of carbon, aluminum, and sulfur concentrations, along with corresponding SEM image, for ALD AlS_x coated NGNS.

Figure 9. Electrochemical performance of ALD AlS_x: (a) charge storage capacity versus number of discharge-charge cycles for ALD NGNS-AlS_x nanocomposites and a mixture of NGNS and micro-sized Al₂S₃, and (b) discharge-charge profiles of the NGNS-AlS_x nanocomposites in the potential window of 0.01 – 3.0 V; (c) charge storage capacity versus number of discharge-charge cycles and (d) discharge-charge profiles of the NGNS-AlS_x nanocomposites in the potential window of 0.6 – 3.5 V.

FIGURES

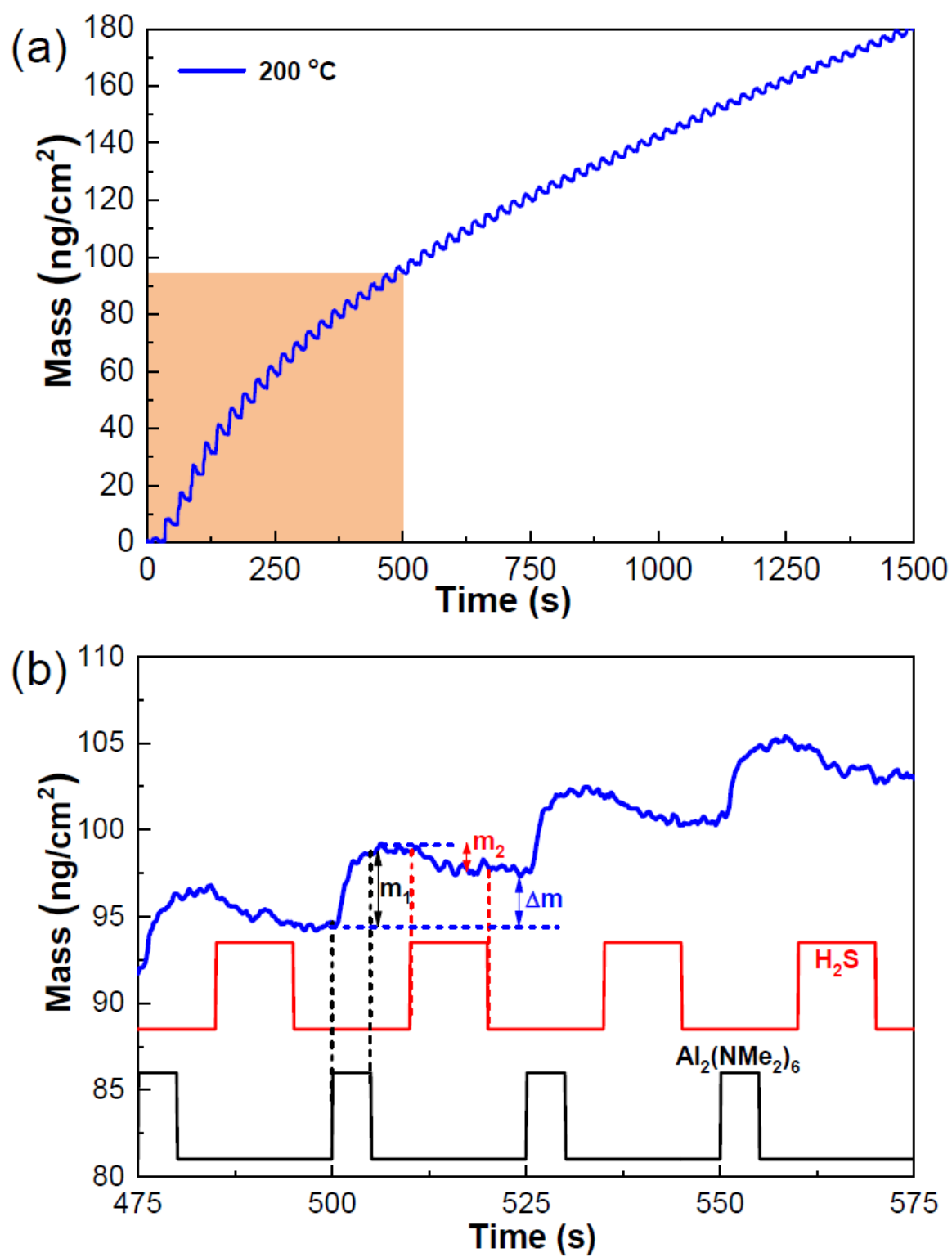


Figure 1.

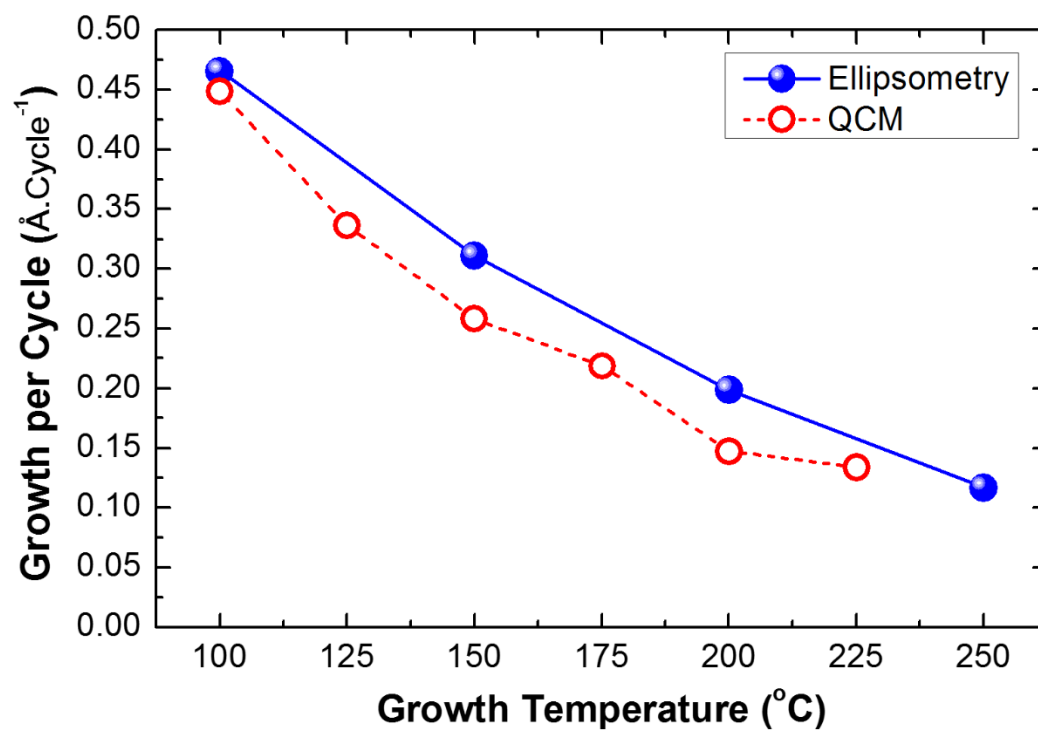


Figure 2.

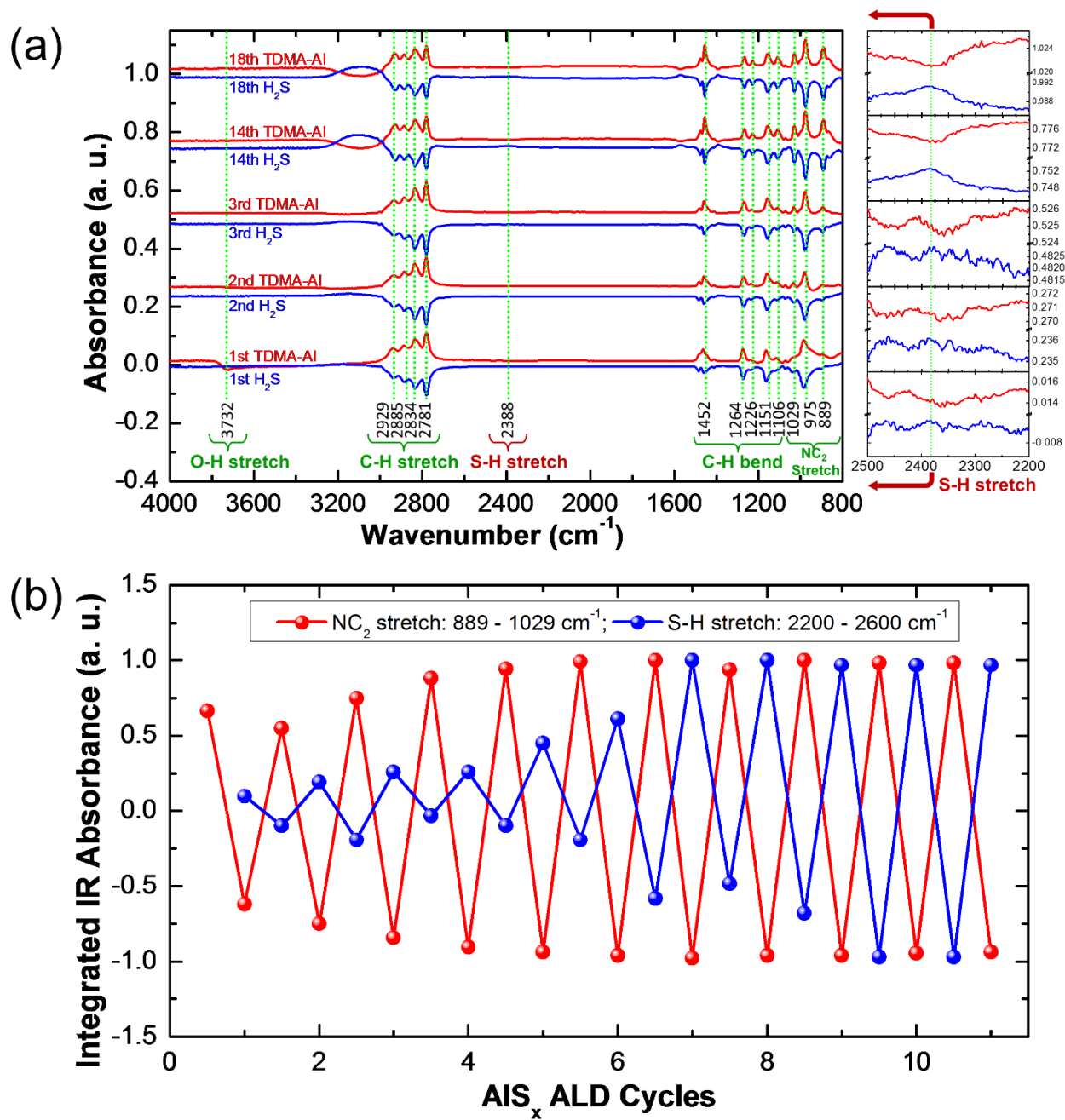


Figure 3.

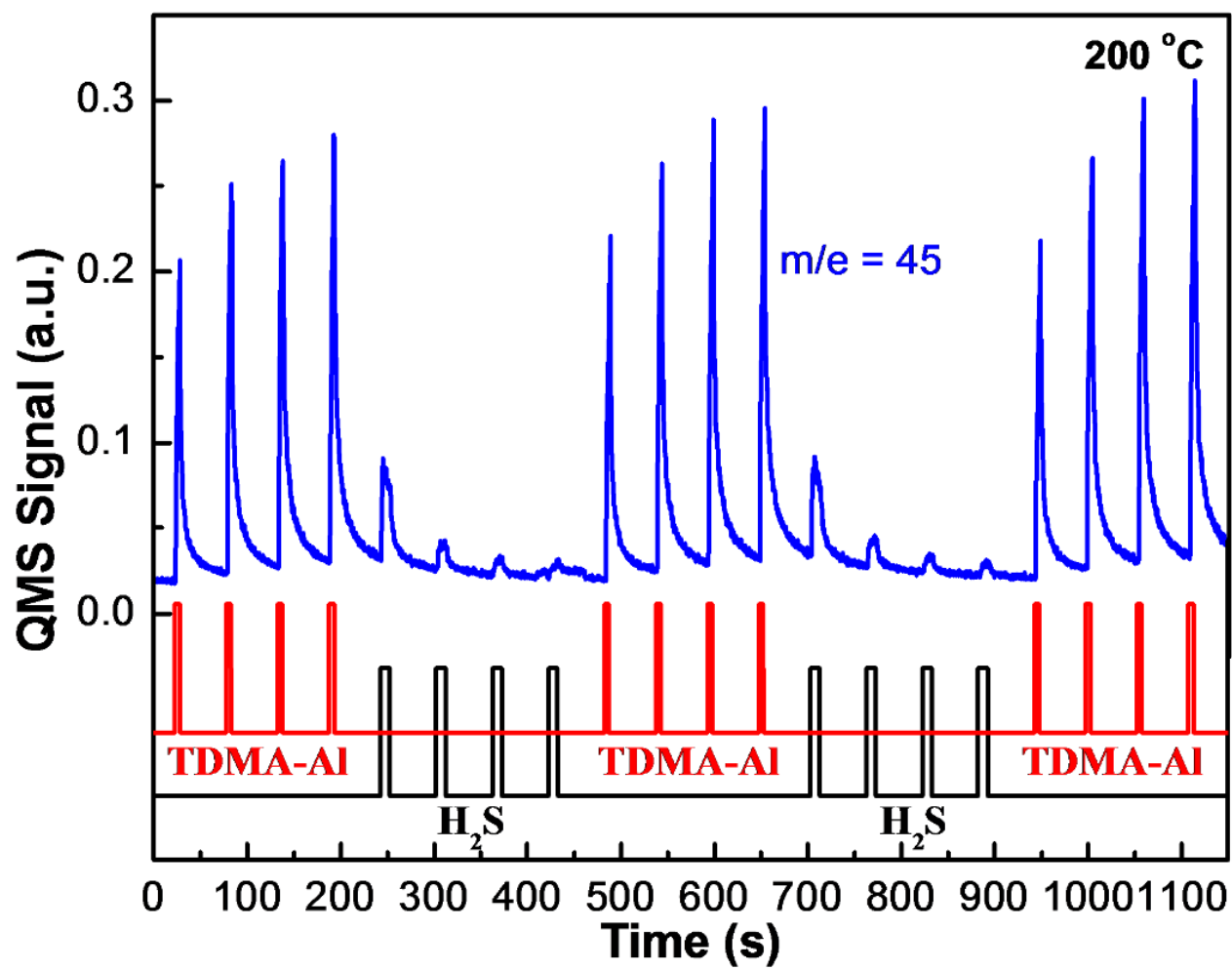


Figure 4.

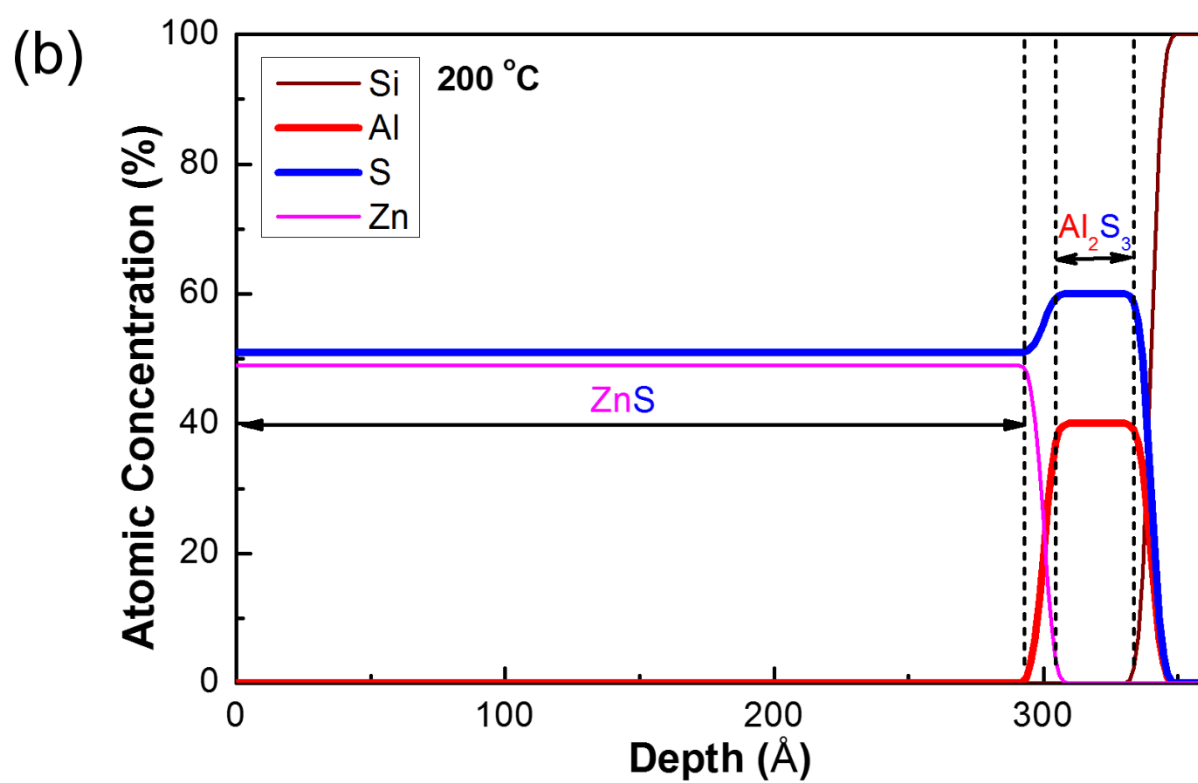
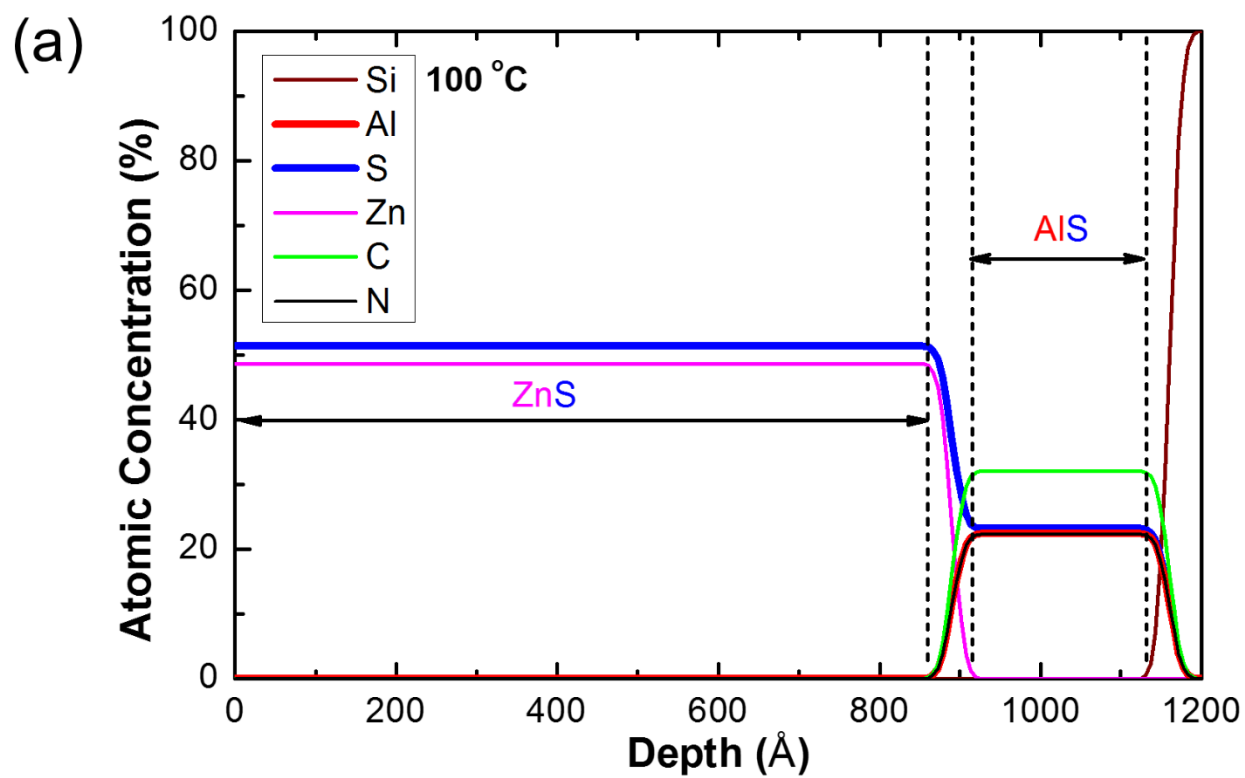


Figure 5.

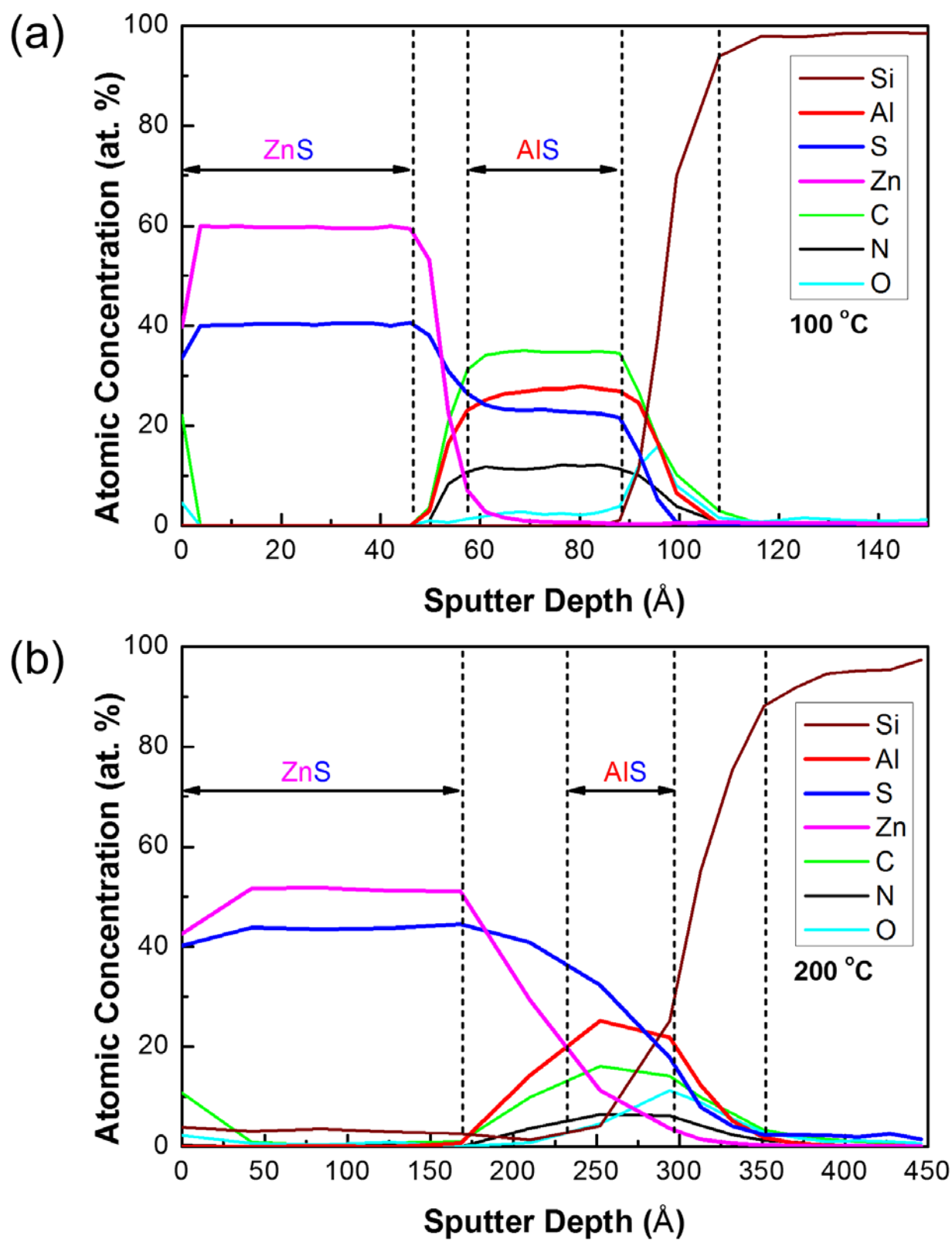


Figure 6.

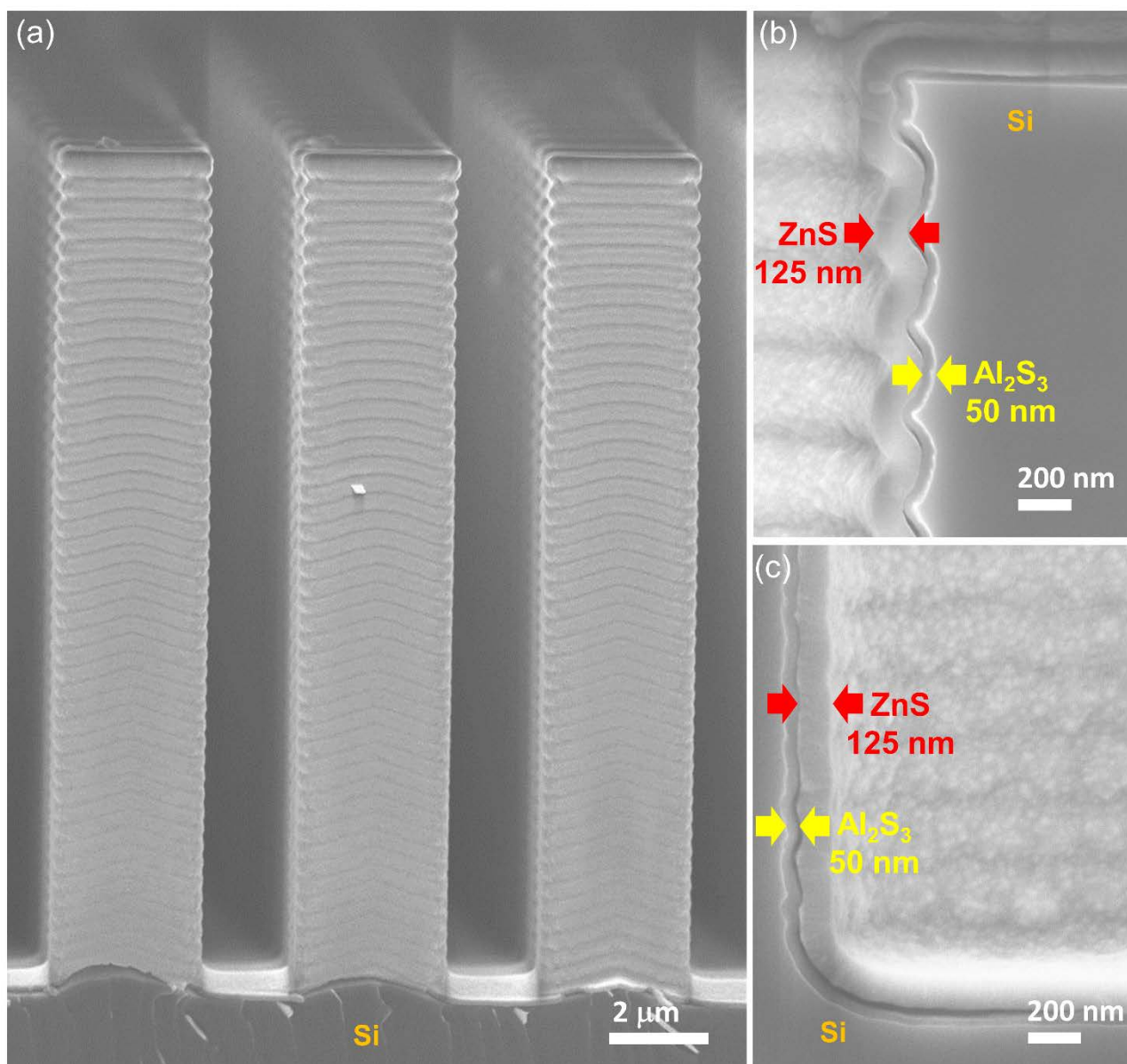


Figure 7.

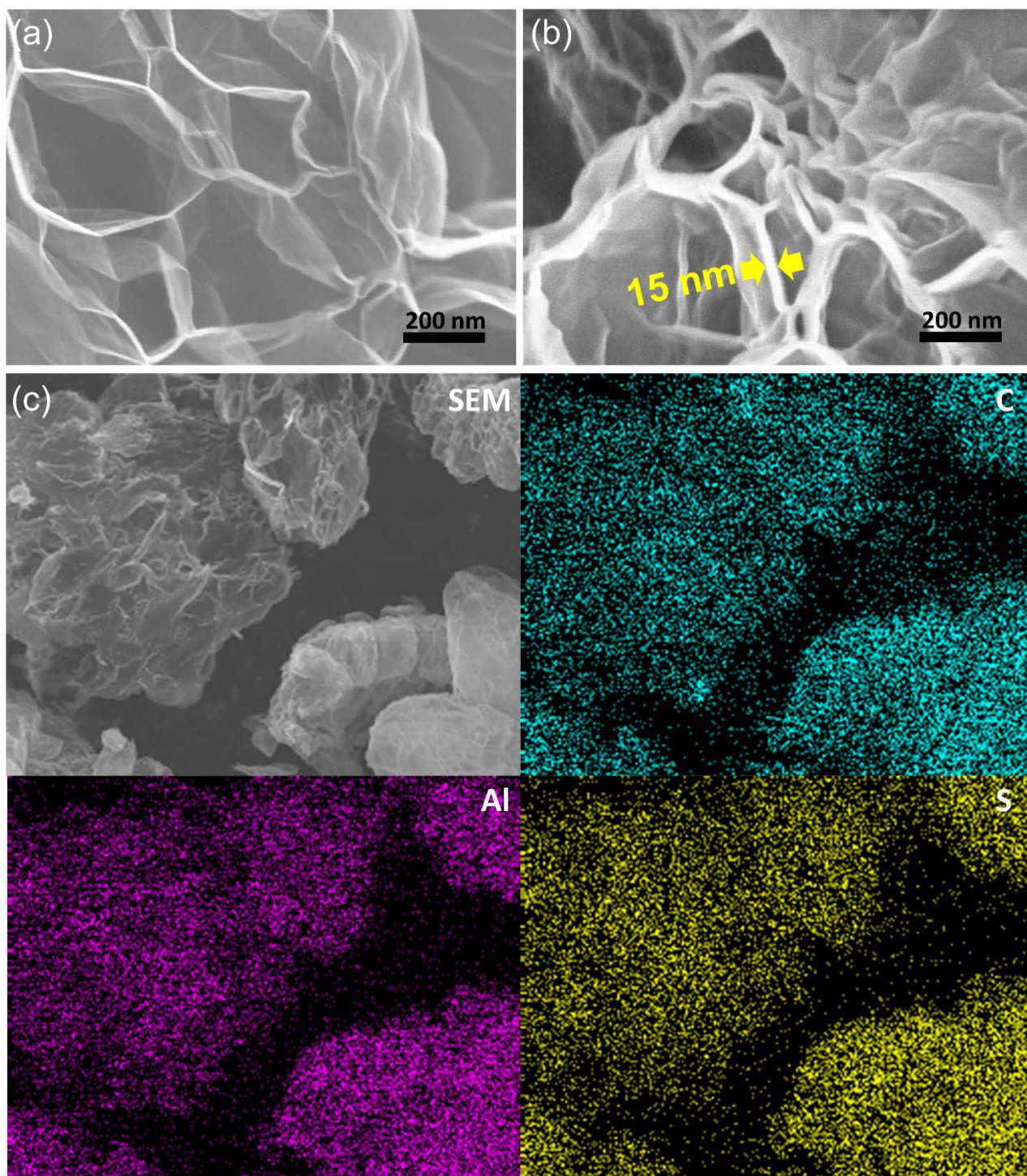


Figure 8.

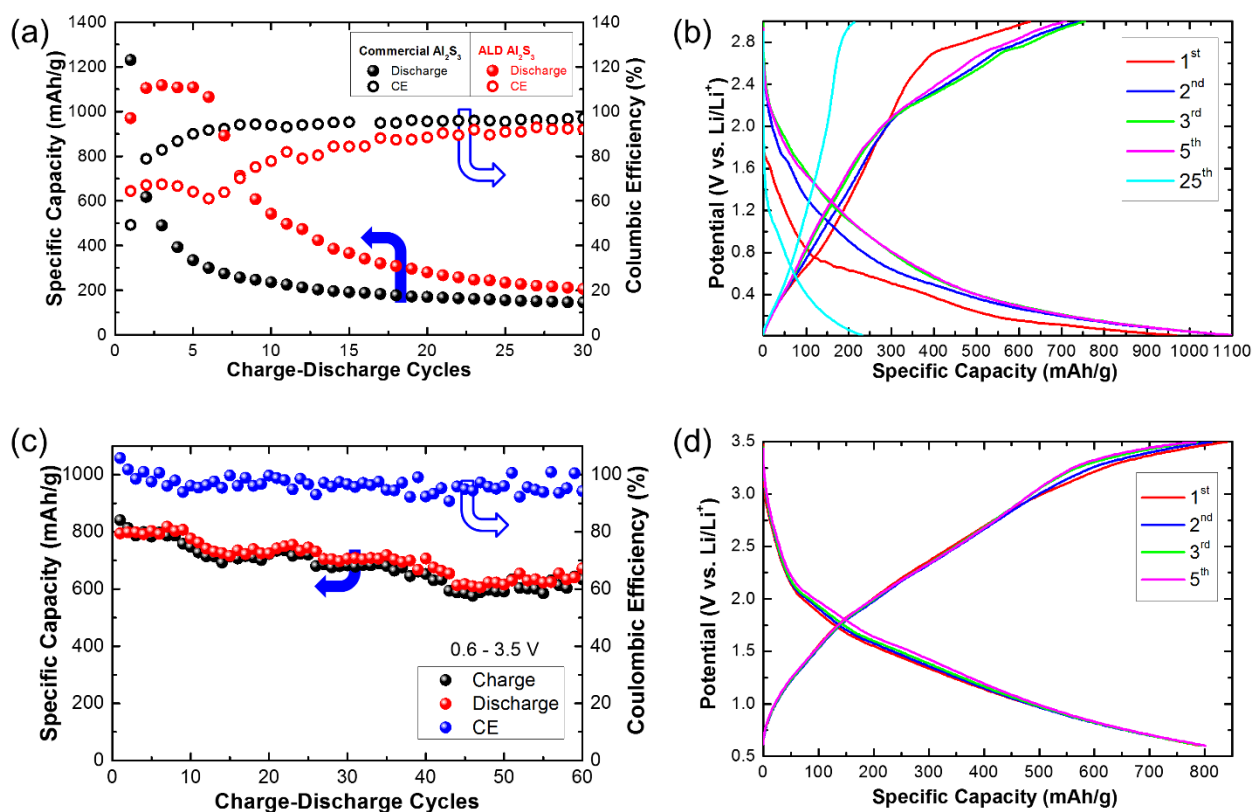


Figure 9.

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